

Pewman Innovation

Footprint of CO2

Industry

Agriculture & Food Systems

Executive Summary

Pewman Innovation is a pioneering agritech deeptech start-up in the field of microbiological and biotechnological solutions aimed at developing climate-resilient and resource-efficient agriculture. Our first two products, CRIOPROTECT and NANOFORTE, are natural biostimulants (and biofertilizers) derived from extremophile microorganisms (from Antarctica and the Atacama Desert) that protect crops from abiotic stress factors such as frost (Crioprotect) and drought (Nanoforte). These solutions reduce irrigation requirements by up to 40%, promote soil regeneration and are 100% organic, offering a sustainable alternative to synthetic agrochemicals. Given the increase in extreme weather events and agricultural risks, Pewman directly addresses the need for climate-resistant crops and stable food supply chains, which closely aligns with LDC's global priorities. Our biostimulants have already proven effective on major crops such as maize (corn), grapes, cereals, and fruit trees, and can be adapted to other key products in LDC's portfolio. In terms of frost protection, CRIOPROTECT allows farmers to avoid traditional fuel-intensive methods (e.g., gas heaters and tractors), significantly reducing CO₂ emissions per hectare. We see Louis Dreyfus Company as a strategic partner in co-developing sustainable value chains, validating our biotechnologies in real-world operations, and scaling up innovative practices that reduce water consumption, chemical inputs and environmental footprint. Together, we can have a measurable impact on the climate, improve farmer productivity and provide traceable, regenerative agricultural solutions at scale.

Pewman Innovation currently spans Latin America and Europe, with production sites in Chile and France and rapidly expanding commercial networks to Switzerland, USA, and other European countries. Supported by public R&D funding and partnerships with strategic agro-industrial players, such as Eurofins (Spain, France, Germany), INRAe (France), Syngenta (Chile), our vision is to catalyse a bioeconomic transformation, combining cutting-edge science with practical solutions for carbon reduction, biodiversity support and long-term agricultural resilience. The Pewman and LDC partnership and collaboration would lead us to cultivating the future of agriculture, from science to stable and large-scale production and supply.

Emissions Overview

The following are estimates of the emissions generated and avoided by the startup - all based on data that the startup entered into the calculator.

Emissions from Operation in 2036

51

ton CO2e

Total Avoided Emissions by 2036

3,569,021

ton CO2e

Scope 1

0

ton CO2e

Scope 2

2

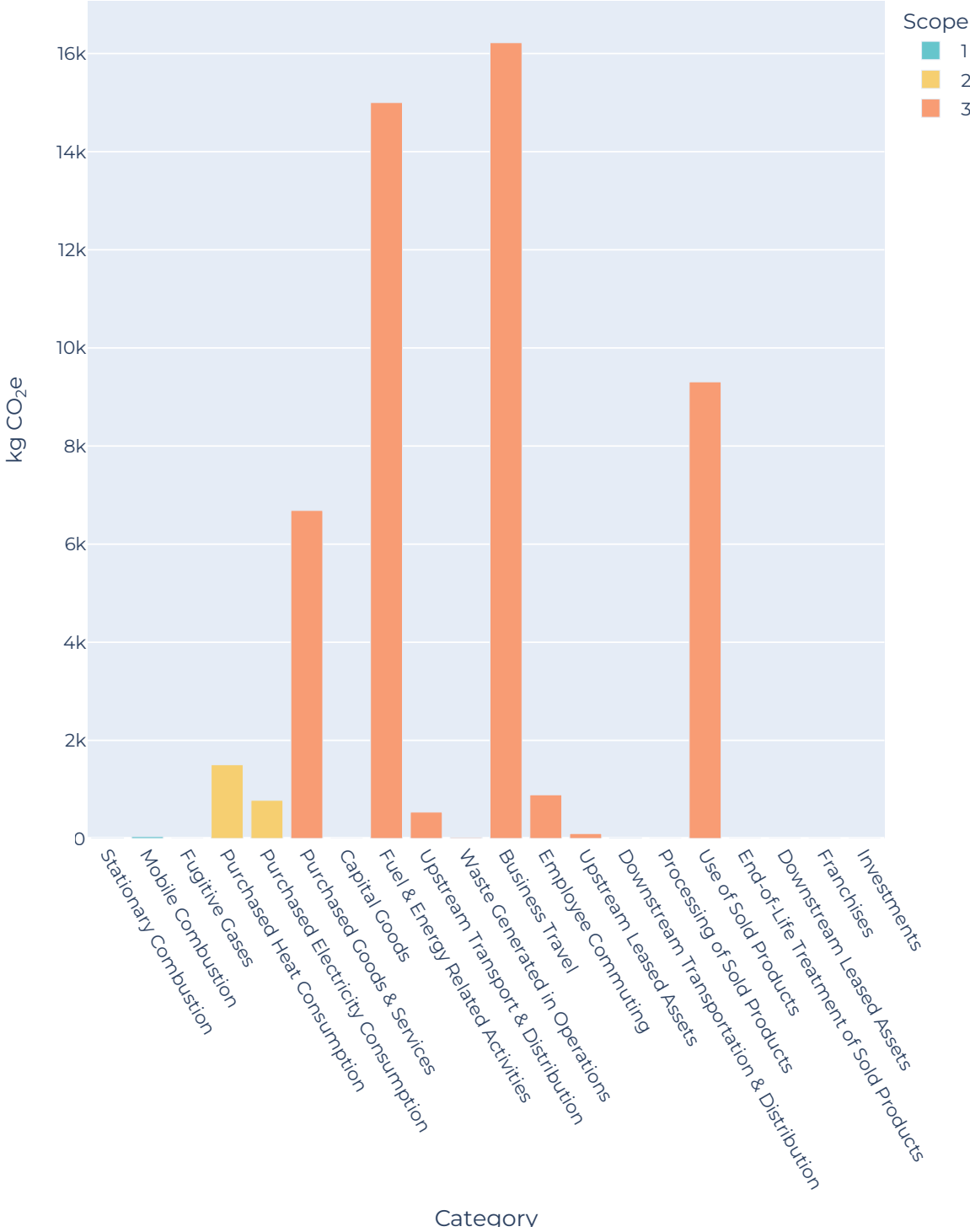
ton CO2e

Scope 3

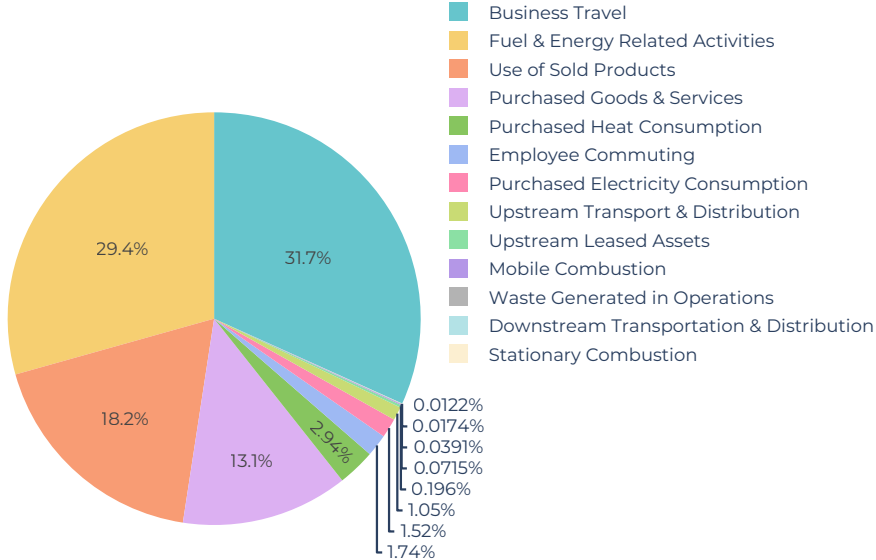
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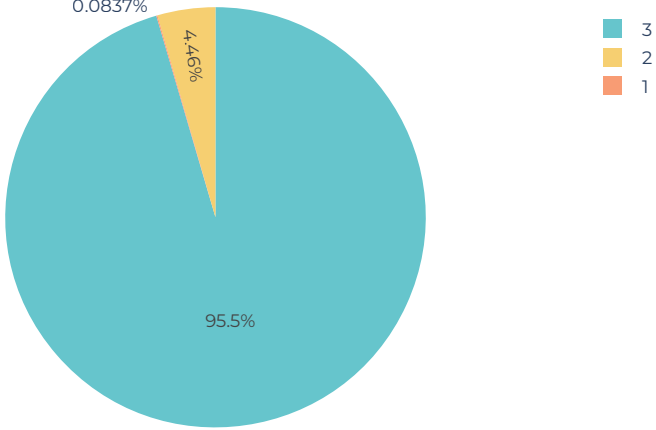
EMISSIONS PER CATEGORY



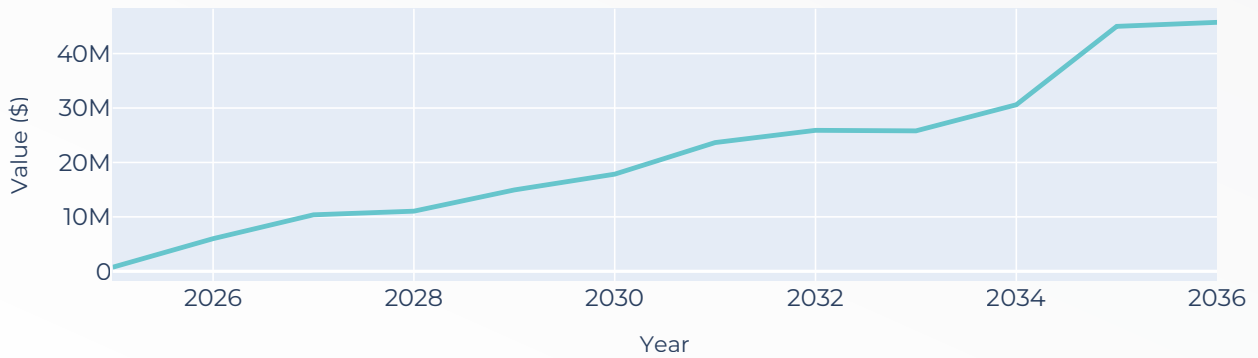
EMISSIONS PER CATEGORY



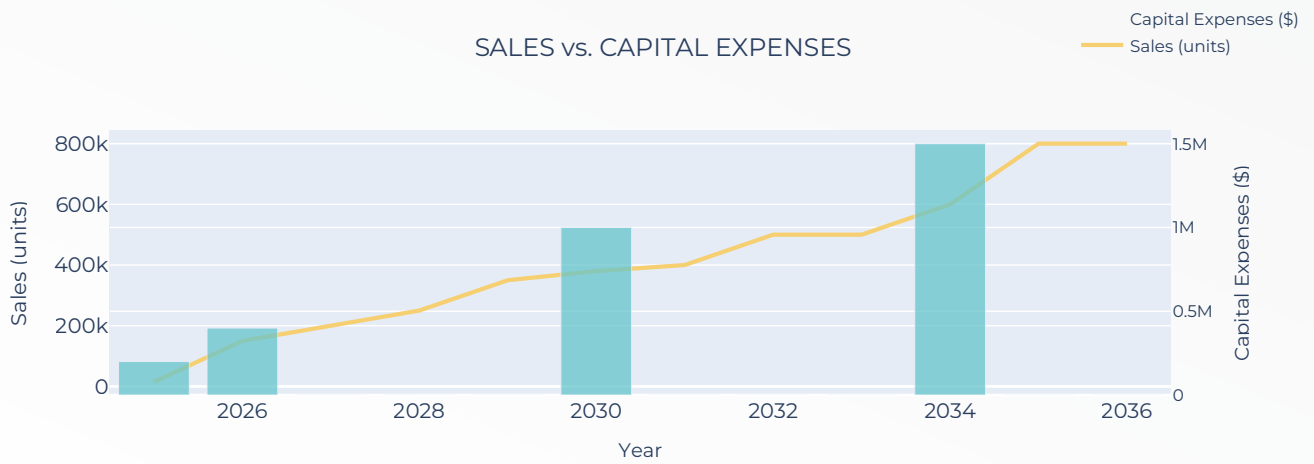
EMISSIONS PER SCOPE



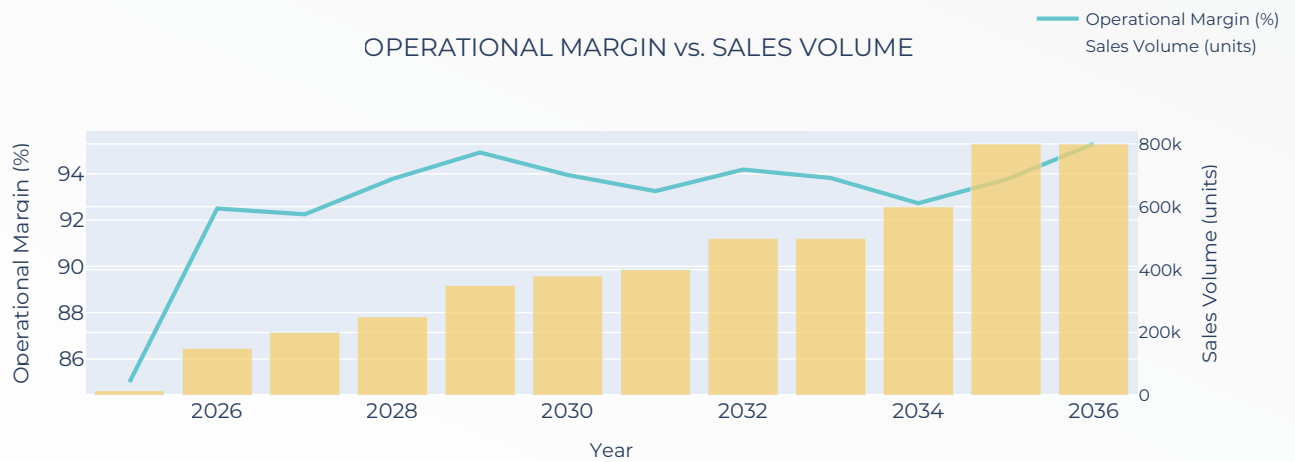
NET CASH FLOW



SALES vs. CAPITAL EXPENSES



OPERATIONAL MARGIN vs. SALES VOLUME

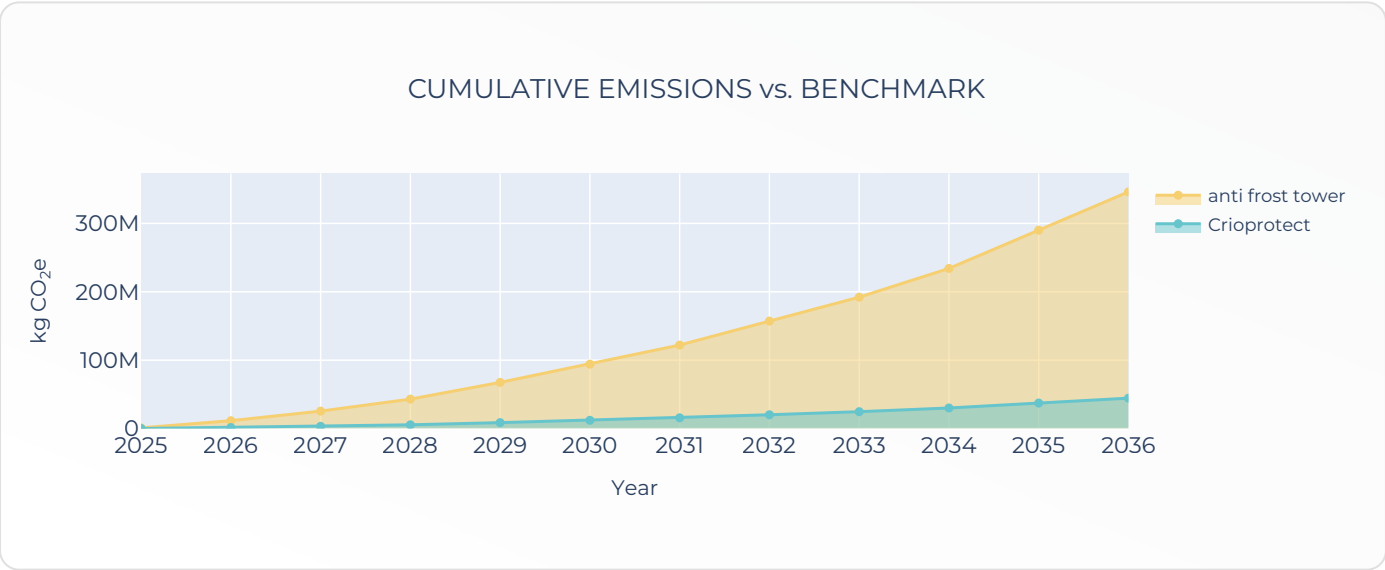


Return on Carbon

The startup did a Return on Carbon analysis using the following frameworks.

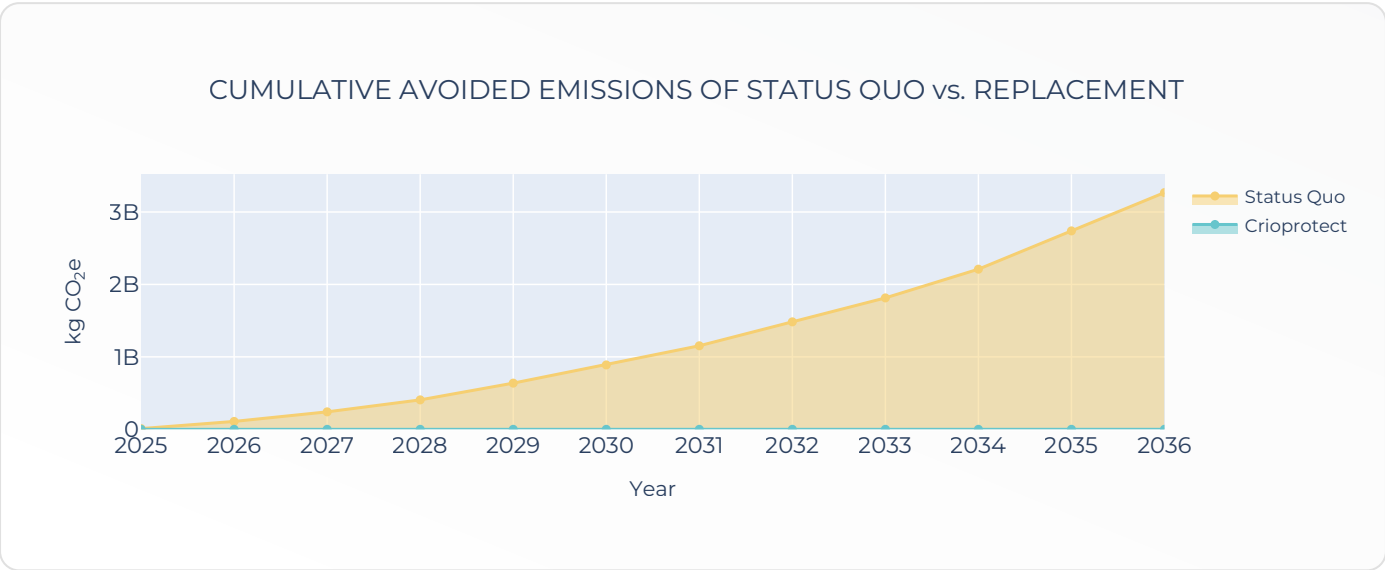
Emissions Avoided through Replacement

The startup compared their product against an existing product that they plan to replace. The comparison is represented at the unit level and cumulatively across time.



Emissions Avoided through Operation

The startup measured the emissions avoided as a result of operational efficiency gains that follow the implementation of their offering. The result is compared to the original operational emissions (Status Quo).



Nature Assessment

The startup has assessed their impact on Nature using the LEAP framework: Locate, Evaluate, Assess and Prepare. The results highlight the impacts and dependencies on nature that the startup deems most impactful, with a justification and an assessment of how they plan to mitigate or capitalize on that impact.

Criterion	Justification	Assessment
Land-use change	Pewman's products focus on enhancing sustainable agricultural practices without promoting land-use change. Their application helps farmers maintain current land usage while improving crop resilience and productivity, thus avoiding any significant shift in land utilization. We support existing ecosystems and contribute to regenerative farming practices, ensuring that agricultural land remains intact, ultimately benefiting soil health and biodiversity while allowing efficient crop management.	We capitalize this low impact by marketing a low impact product.
Freshwater-use change	Our product doesn't impact on the amount of freshwater used by farmers. We promote responsible freshwater management in agricultural operations.	Pewman's CRIOPROTECT and NANOFORTE products exhibit low impact on freshwater use. They are designed to optimize irrigation efficiency, allowing farmers to save up to 40% of water compared to conventional practices. This reduction not only conserves water resources but also minimizes competition for freshwater in agricultural settings. By facilitating sustainable farming practices, Pewman's innovative solutions support water conservation efforts while enhancing crop productivity.
Ocean-use change	Our product does not impact ocean use	We do not capitalize this aspect and do not plan to capitalize on it.

Criterion	Justification	Assessment
Greenhouse gas emissions	All current frost protection alternatives are energy-intensive (wind turbines, sprinklers, outdoor heaters, etc.), so our product has very low greenhouse gas emissions in comparison.	We are evaluating an alliance to capitalize on carbon credits to those farmers who change their current method for our product. Also, fuel in Chile is expensive, so we use this criterion as a marketing strategy.
Water use	Our product use a minimal part of the water use of a farmer's product application programs, usually consisting several dozens of products. Also, the product can be mixed with other products in the same tank, further reducing a change in the water use.	We capitalize this low impact by comparing our products to other water-intensive methods of frost protection, as sprinklers.
Other resource use	Low usage due to sustainable sourcing and practices.	We do not capitalize this criterion. We plan to capitalize it by obtaining different national and global certifications of low impact.
Non-GHG air pollution	The use of our product doesn't emit any non-GHG air pollution.	We do not capitalize this aspect and do not plan to capitalize on it.
Water pollution	Our product is fully biodegradable and 100% organic, minimizing water pollution. We already possess the ECOCERT for both products in Europe.	We do not capitalize this aspect. We can associate with responsible use of water initiatives.
Soil pollution	Our product is fully biodegradable and 100% organic, minimizing soil pollution.	We do not capitalize this aspect. We can associate with responsible use of soil initiatives.
Waste	Our operations generate minimum waste.	We do not capitalize this criterion. We plan to make alliances to reduce the plastic waste of our clients once they use the product and dispose the bottles.
Disturbances (Light & Noise)	Our solution can replace Wind turbines and helicopters, which usually generate noise	

Criterion	Justification	Assessment
	contamination, affecting neighbors.	We capitalize this as part of our marketing strategies against other alternatives.
Biological alterations	We declared a medium impact because we introduce a new species of bacteria, although soil genomic studies show little to no disruption of the native microbiome. The product doesn't affect any other animal, plant or insect.	We capitalize this as part of our marketing strategy as a safe, 100% organic and biodegradable solution.
Extent	Our impact on the extent of the environment is minimal due to the nature of the solution.	We do not capitalize on this criterion. We can capitalize it by alliances with environmental impact initiatives.
Condition	Careful management practices as the use of our product ensure that the condition of agricultural land stays strong and productive.	We can capitalize this by offering soil condition analysis to demonstrate the safety and sustainability of our solution.
Population size	Current operations have little impact on populations.	We do not capitalize on this criterion and we don't plan to capitalize on it.
Extinction risk	The use of our product employ practices that do not threaten local species.	We can capitalize this by promoting biodiversity-friendly practices to enhance local flora and fauna populations
Biomass provisioning	Our product doesn't impact this criterion in any significant way.	We do not capitalize on this criterion.
Water supply	Medium usage levels indicate a balanced approach to water resources, maintaining ecosystem health.	Invest in efficient water management technologies to optimize both crop and wildlife needs.
Water purification / flow regulation / maintenance	Medium evaluations suggest an opportunity to enhance water purification through better management practices.	Implement buffer zones and filters to enhance water quality further and support aquatic ecosystems.
Soil quality regulation / soil and sediment	High scores indicate significant positive impacts on soil quality	Continue current practices and educate local farmers on soil

Criterion	Justification	Assessment
retention / solid waste remediation	and retention due to best practices in agriculture.	management to increase the overall quality of agricultural soils.
Pollinations / pest and disease control / nursery population / habitat maintenance	High impact; agricultural practices actively support pollinators and pest control through integrated pest management and habitat creation.	Collaborate with local pollinator protection programs and plant native species to enhance habitats that promote pest control and pollination services, benefiting biodiversity.
Flood or storm mitigation / noise attenuation	High impact; practices contribute to climate resilience	Explore resilience of crops implement carbon capture strategies as part of a climate-positive strategy.
Global or local climate regulation / rainfall pattern regulation / air filtration	High impact; practices contribute to carbon sequestration and climate regulation, aiding in resilience against climate change effects.	Explore reforestation initiatives and implement carbon capture strategies as part of a climate-positive strategy, enhancing climate regulation efforts.
Recreation / visual amenity / scientific and education / spiritual, artistic, symbolic	High impact in scientific and educational activities in farmers, schools and STEM areas	Increase the number of the activities with impact in society

Assumptions

The startup has made the following assumptions to back up their claims.

Global Assumptions

- These global assumptions provide a comprehensive framework for evaluating the environmental impact of Pewman Innovation's products and the overall sustainability of its business operations in the context of the MIC calculations:
- 1. Sustainability Integration: All products (CRIOPROTECT and NANOFORTE) are designed with sustainability in mind, aiming to reduce the environmental impact of agricultural practices and promote resource efficiency.
- 2. Complete Carbon Footprint Assessment: The GHG emissions will be calculated comprehensively, taking into account direct (Scope 1), indirect (Scope 2), and value chain (Scope 3) emissions associated with operations, product usage, and waste management.
- 3. Market Adoption: We assume a gradual market adoption of our products, with increasing acceptance and application rates in both existing markets (Chile and Peru) and new markets as we expand into Europe.
- 4. Long-Term Benefits: The benefits of carbon reduction, water savings, and operational efficiency will accumulate over time as more farmers implement our products and regenerative practices, leading to significant long-term environmental impacts.
- 5. Partnership and Collaboration: Strategic partnerships with distributors, regulatory bodies, and agricultural organizations will provide support for successful product deployment and regulatory compliance in new markets.
- 6. R&D Commitment: Continuous research and development will enable us to innovate further, improve existing products, and develop new formulations that enhance sustainability and carbon sequestration.
- 7. Climate Resilience: Our products are expected to contribute to the climate resilience of agricultural practices, helping farmers adapt to changing conditions, such as droughts, frosts, and extreme weather events.
- 8. Data Collection for Impact Measurement: We will implement robust systems for monitoring and reporting the impact of our products on carbon emissions, water usage, and soil health, ensuring transparency and accountability in our environmental claims.
- 9. Regulatory Compliance: All operations, products, and innovations will adhere to applicable regulations and standards in both local and international markets, particularly with regard to environmental sustainability and agricultural safety.
- 10. Impact of Innovation on Agriculture: By replacing conventional methods with our innovative solutions, we will contribute to the broader goal of sustainable agriculture, reducing reliance on chemical inputs, minimizing waste, and maintaining ecosystem health.

Business Forecast

Capital Activities

→ We have recently established a new facility focused on research and development (R&D) as well as bioproduction, utilizing a credit secured from the BID (Banco Interamericano de Desarrollo) in Santiago, Chile. Additionally, we have opened a new branch in France and acquired 200k euros in non-dilutive funding to kickstart a bioproduction unit in Alsace, alongside an established partnership in Strasbourg. We plan to supplement this capital with an additional 0.5 million euros by the end of 2026 in France, with a funding request already submitted to BPI France. Subsequently, we expect growth in our sales and aim to raise an initial round of funding amounting to 3 million euros by 2027.

Sales

→ Sales for 2026, 2027, and 2028 will be driven by strong partnerships in both Latin America and Europe. In Chile, we have secured a sales contract with our distributor CALS running until 2027, and expansion of sales in Chile and into Peru is confirmed for 2026-2027, in collaboration with Syngenta. In Europe, we are currently finalizing negotiations with two major agricultural distribution partners covering France, Germany, Belgium, and Spain, supported by ongoing regulatory registration processes in collaboration with Eurofins, Syngenta, and CertisBelchim. These partnerships and the favorable positioning of our products (CRIOPROTECT and NANOFORTE) in climate-resilient, organic, and regenerative agriculture will be key drivers of growth.

→ In parallel, we are preparing the next phase of expansion, including the deployment of new R&D and production facilities across South America, North America, and Central Europe. This strategic infrastructure, supported by incubators, public innovation programs, and industry partners, will enable us to scale production, ensure local adaptability, and accelerate market penetration.

→ Given these developments, a strategic partnership with Louis Dreyfus Company (LDC) would represent a high-impact opportunity. LDC's global reach and strong presence in major crop value chains such as maize, wheat, soybean, and sugarcane aligns with our focus on abiotic stress resilience (e.g., drought, frost), key pain points for large-scale producers. Our technologies can help reduce water and fertilizer inputs, improve supply stability, and support LDC's sustainability goals by offering low-carbon, nature-based solutions. Furthermore, co-developing field trials and deployment models within LDC's sourcing networks could unlock synergies in climate adaptation, supply chain traceability, and regenerative transition for key crops.

→ Together, Pewman Innovation and LDC could reshape input strategies in climate-vulnerable regions, enabling farmers to transition away from resource-intensive practices, while ensuring secure and resilient supply for agri-commodities critical to LDC's operations.

Unit Price

→ The unit price is set at 40 USD for the first three years. Following this period, a slight increase is expected for both products over the next seven years, reaching up to 60 USD. This adjustment is influenced by the current market trends in biostimulants and their projected growth, as well as rising costs of raw materials and transportation. The pricing is based per liter of product, with an application rate of 2 to 5 liters per hectare, depending on the crop, and a maximum of 2 to 3 applications per season per He.

Funding & Other Income

→ In 2025, we obtained a financing credit from the BID (Banco Interamericano de Desarrollo) amounting to 180,000 USD. Looking ahead to 2028, we will require additional investment for our European expansion. We anticipate that fundraising will be necessary to secure 3,000,000 USD to support these growth initiatives. We have recently established a new facility focused on research and development (R&D) as well as bioproduction, utilizing the credit secured from the BID in Santiago, Chile. Additionally, we have opened a new branch in France and secured 200k euros in non-dilutive funding to kickstart a bioproduction unit in Alsace, alongside an established partnership in Strasbourg. We plan to supplement this capital with an additional 0.5 million euros by the end of 2026 in France, with a funding request already submitted to BPI France. With these funding initiatives, we aim to support our anticipated sales growth and secure further investments to enhance our operations.

COGS

- Biostimulant Production (Non-exhaustive List)
- Raw Materials:
 - -Culture media and fermentation nutrients
 - -Extremophile microorganisms (starter strains)
 - -Stabilizers (natural polymers)
 - -Water (non-purified/ and laboratory grade-purified)
- Primary Packaging:
 - -Plastic bottles (various sizes)
 - -Caps, seals, droppers or dosing accessories
 - -Labels (printed or adhesive)
 - -Outer cartons or trays
- Secondary Packaging:
 - -Boxes for shipping
 - -Pallets, shrink-wrap, buffers or void-fill materials
- Manufacturing & Formulation Costs:
 - -Bioreactor operation (energy, maintenance):
 - -Fermentation and culture processing
 - -Downstream processing (centrifugation, filtration, stabilization)
 - -Dilution and concentration steps
 - -Quality control and microbiological testing
 - -Labor costs related to production and packaging
- Logistics & Transportation:
 - -Refrigerated or ambient transportation (domestic or international)
 - -Warehousing or temporary storage
 - -Customs/duties for export markets (Chile → EU, etc.)
 - -Last-mile delivery to distributors/farmers
- Consumables & Supplies:

- -Laboratory disposables (pipettes, flasks, tubes, gloves)
- -Cleaning agents and sanitizers
- -Safety and protective equipment for staff
- -CO₂, nitrogen, and other gases
- Traceability & Compliance:
- -Batch tracking systems
- -Certification costs (organic, ISO, etc.)
- Product Losses / Waste Management:
- -Rejected batches
- -Expired stock
- -Cleaning and effluent disposal

Fixed Costs

- -Salaries & Social Charges
- • Full-time R&D staff (engineers, scientists, etc)
- • Production team
- • Administrative & management staff
- • Interns or assistants
- -Rent & Facility Charges
- • Office and laboratory rent (Santiago, Chile; Strasbourg, France; etc.)
- • Utilities (electricity, water, heating, internet)
- • Facility services (cleaning, security, access)
- -Depreciation & Equipment Leasing
- • Bioreactors and fermentation systems
- • Lab equipment (centrifuges, incubators, pipettes)
- • Computers, office hardware, etc.
- • Leasing of lab or production equipment
- -Equipment Maintenance & Calibration
- • Scheduled maintenance contracts (for reactors, pumps, cold rooms)
- • Calibration services for lab equipment
- • Spare parts, service fees
- -Insurance
- • Liability insurance (product and professional)
- • Equipment/laboratory insurance
- • Health and employment-related insurance contributions
- -Professional Services
- • Legal and IP counsel (e.g., Inspire Avocats, Achaka Conseils)
- • Regulatory consultants (e.g., Eurofins, RITTMO)
- • Accounting & audit services

- • Administrative or HR support (outsourced)
- • Software & Licenses
- • Lab software (e.g., for traceability or batch control)
- • Office suite, CRM, data storage
- • Specialized biotech analysis tools (e.g., bioinformatics)
- • Communication & Marketing
- • Website hosting and maintenance
- • Promotional materials and content creation
- • Attendance at fairs and scientific events (fixed budgeted cost)
- • Training & Team Development
- • Continuous learning, conferences, safety certifications
- • Onboarding and staff development resources

Operation Activities

Scope 1

- These assumptions were made using average emission factors that can be considered equivalent for producing our products in Chile or in Europe (where the first two production and R&D labs of Pewman Innovation are located).
- 1. Drinking water -> For producing 1 liter of product we use a liter of water. -> 30 kg
- 2. Peptides -> For preparing the medium to grow the bacteria, we use tryptone. We assumed it to be a generic peptide since it is digested casein. -> For 1 liter of medium, we use 0,01 kg of tryptone, which means that for 1 liter of the final product, we use 0,00001 kilograms of tryptone. To protect 1 hectare, we multiply this value by 30 -> 0,0003 kg
- 3. Sodium Chloride -> For preparing the medium to grow the bacteria, we use NaCl. -> For 1 liter of medium, we use 0,01 kg, which means that for 1 liter of the final product, we use 0,00001 kg of NaCl. To protect 1 hectare, we multiply this value by 30 -> 0,0003 kg
- 4. Yeast extract -> For preparing the medium to grow the bacteria, we use yeast extract. Due to the lack of specific data, we assumed that the yeast extract has the same environmental impact in its life cycle assessment (LCA) as regular yeast. (<https://www.afl evadura.com/sostenibilidad/huella-de-carbono>) -> For 1 liter of medium, we use 0,005 kg, which means that for 1 liter of the final product, we use 0,000005 kg of yeast extract. To protect 1 hectare, we multiply this value by 30 -> 0,00015 kg
- 5. PET -> We use PET for our packaging, specifically for the containers. For 30 liters of product, one 20-liter container and two 5-liter containers can be used.-> We weighed the containers to calculate the total PET used to protect 1 hectare. -> 1,357 kg
- 6. Glycerol -> The impact of glycerol was not found in the database, so we referenced it (<https://apps.carboncloud.com/climatehub/product-reports/id/16243501676>). Glycerol is used to cultivate the bacteria. -> For 1 liter of bacteria, we use 0,0025 kg of glycerol, which means that for 1 liter of the final product, we use 0,0000025 kg. To protect 1 hectare, we multiply this value by 30 -> 0,000075 kg.
- For the transport we consider an average client that is located 100 km from the production center.
- 7. Truck (Crioprotect) -> to protect one hectare per year we need to transport 30 kg (0,03 ton) of product though 100 km. -> 3 tkm

Scope 2

- Electricity Chile -> For the production of the product, we cultivate 4 liters of concentrated bacteria in a bioreactor for 24 hours. The bioreactor model used is the Minifors 2 (Infors AG). According to the manual, under the Connection Requirements section, the maximum power consumption is listed as 800 W -> First, we calculated the energy required to produce 4 liters of concentrate by multiplying the maximum power consumption by the hours of use. This value was then divided by 4 and subsequently by 1000 (since our product concentration is 0,1%) to determine the energy needed for just 1 liter of the final product. Finally, the resulting number was multiplied by 30 to calculate the energy used to protect 1 hectare -> 0,144 kWh
- For Europe → These assumptions were made using an average amount of electricity and equivalent factors applied to the production of our products in Chile and in Europe (where the

first two production and R&D labs of Pewman Innovation are located). The two values were then added here.

Scope 3

- 1. We currently consider 10 full time employees in Chile and 3 full time employees in France, 5 USD per serving, 2 servings a day during work days, 20 days a month, 12 months a year.
- 2. Capital goods are left blank as asked.
- 3. We are not sure which fuel and energy related activities are linked to our case.
- 4. We considered the transport of the raw materials and plastic bottles, with the amount being how much we approximately spend in a year in each case.
- 5. There are no significant waste products generated by the production of our product (Bioreactor to bottled product)
- 6. For Business travel, we spend around 4000 USD a year to attend different meetings and events nationally and around 12000 USD internationally
- 7. We considered a 60km/day round trip for employee commuting, with a 15km/liter yield and 1,5 USD/liter cost that results in 6 USD/day, then multiplied this by 20 days a month, 12 months a year and for 3 cars in total.
- 8. We don't lease assets
- 9. We considered the product delivery, which is around 5 tons per order, to the distributors warehouse, which is around 20km from our manufacturing site.
- 10. Either us nor our clients process the sold products in any way.
- 11. We considered the use of the product by the farmer as the tractor covering around 3 Km per hectare (several up and down paths between crops) and dragging 2 tons at the start and 0 tons at the finishing point, so 1 ton average, giving us a total of 3 tons*Km. Also, we use 20-liter bottles that weigh around 1 Kg when empty (including the bottle cap). Each year we sell around 3000 20-liter bottles. It is not a waste of our production site, but it is a waste generated immediately by the use of our product.
- 12. Our clients don't usually apply any end-of-life treatment to the sold product.
- 13. We don't lease assets downstream
- 14. We do not have any additional franchises
- 15. Our company does not have any investments yet.

Return on Carbon

Emissions Avoided through Replacement

- 1. Product Replacement: Crioprotect is intended to replace traditional anti-frost towers used in the field for protecting sensitive crops from frost damage.
 - 2. Emissions from Anti-Frost Towers: Anti-frost towers typically consume significant electricity or fossil fuels to function, leading to higher greenhouse gas emissions associated with their operation.
 - 3. Efficacy of Crioprotect: Crioprotect provides effective frost protection while significantly reducing the need for energy-intensive frost prevention methods, resulting in lowered emissions.
 - 4. Application Rate: Crioprotect is applied at an average rate of 2 to 5 liters per hectare, depending on the crop type and growing conditions.
 - 5. Reduction in Energy Use: By replacing anti-frost towers with Crioprotect, it is estimated that energy use can be reduced by approximately 70-90%, translating to lower GHG emissions from energy consumption.
 - 6. Timeline for Replacement: The transition to Crioprotect will occur gradually over the next several years as farmers adopt the new technology alongside existing methods.
 - 7. Carbon Offset: The use of Crioprotect is expected to contribute to carbon offset programs through the reduction of fossil fuel use and associated emissions.
- Replacement of a Product (ROC) for Anti-Frost Towers:
- - Current Product: Anti-frost towers
 - - Claimed Footprint: Each anti-frost tower operation generates approximately 70 kg CO₂e per hectare per frost event.
 - Replacement Product: Crioprotect
 - - Claimed Footprint of Crioprotect: The application of Crioprotect results in estimated emissions of only 10 kg CO₂e per hectare per frost event due to reduced energy usage and avoidance of traditional frost prevention methods.
 - Emissions Reduction: The implementation of Crioprotect in place of anti-frost towers leads to an avoided emissions footprint of 60 kg CO₂e per hectare per frost event.
 - Projected Adoption Rate: It is anticipated that adoption of Crioprotect will reach 30% of the frost-prone crop area in the first few years, increasing as awareness and efficacy demonstrate clear benefits.
- For our Crioprotect product, glycerol can be replaced by 'raw glycerol' by 2030, which is a waste product in the biodiesel industry, thus reducing environmental impact. Plastic bottles will be recovered and recycled as part of a circular economy approach. Transportation can be conducted using electric trucks to lower carbon emissions.
 - Furthermore, employee travel will shift towards more sustainable transportation methods, emphasizing reduced flights and increased use of trains for conferences and events.
 - We also plan to implement a robust tracking system to monitor transportation emissions, ensuring continuous improvement towards sustainability goals.
 - Additionally, we are exploring partnerships with organizations that provide carbon offset programs to further mitigate our environmental impact.
 - The application of Crioprotect leads to avoided emissions in agricultural practices (like reduced fuel usage from frost protection).

Emissions Avoided through Operation

→ Our innovations, CRIOPROTECT and NANOFORTE, lead to less fuel and water usage in agricultural practices, resulting in reduced operational costs and environmental impacts.

Specifically, we expect the following assumptions:

→ Reduced Water Usage: CRIOPROTECT helps in reducing water needs for frost protection, which in turn minimizes the overall water consumption per hectare of crops. NANOFORTE optimizes irrigation, allowing farmers to save up to 40% of water, which is critical in water-scarce regions.

→ Less Fuel Consumption: By replacing energy-intensive frost prevention methods (such as anti-frost towers) with CRIOPROTECT, we anticipate a significant reduction in fuel usage. This translates to lower GHG emissions from fossil fuel combustion.

→ Operational Efficiency: The use of our products is expected to streamline agricultural operations by minimizing the need for additional inputs like chemical fertilizers, which often require significant energy and resources to produce and apply.

→ Enhanced Crop Resilience: Our products not only contribute to resource efficiency, but also boost crop resilience against environmental stressors, potentially leading to fewer resources needed for recovery and maintenance of crops.

→ Long-Term Sustainability: Overall, the adoption of CRIOPROTECT and NANOFORTE is expected to shift agricultural practices toward greater sustainability, fostering a shift to resource-efficient farming that ultimately benefits the environment and reduces operational emissions.

→ These assumptions demonstrate our commitment to driving operational efficiencies and achieving measurable reductions in carbon emissions through our innovative biostimulants.

Sequestered Emissions

→ These assumptions reflect our understanding of the potential for carbon sequestration associated with our products, while recognizing the complexities of precise calculations:

→ -Product Application Impact: We expect that using Crioprotect and NANOFORTE will enhance soil health, leading to increased carbon storage in the soil.

→ -Improvement in Soil Microbial Activity: Our products are designed to boost microbial activity in the soil, which helps in retaining more carbon.

→ -Adoption of Regenerative Practices: We assume that farmers who use our products will also adopt practices like cover cropping and reduced tillage that support carbon storage.

→ -Long-Term Carbon Retention: We believe that the increased carbon captured in the soil will be held over time, although exact retention rates may vary.

→ -Climate Resilience Benefits: By helping crops withstand climate stresses, our products could lead to more stable yields, reducing the need for extra inputs and promoting sustainability.

→ -Ecosystem Enhancements: Improved soil health will likely benefit biodiversity, which plays a role in supporting long-term carbon sequestration.

→ -Monitoring Measurement: We acknowledge that calculating specific sequestered emissions per product per year is currently challenging.

→ Thus, we are focusing on establishing methodologies to measure changes in soil carbon over time but will not specify exact values at this stage.